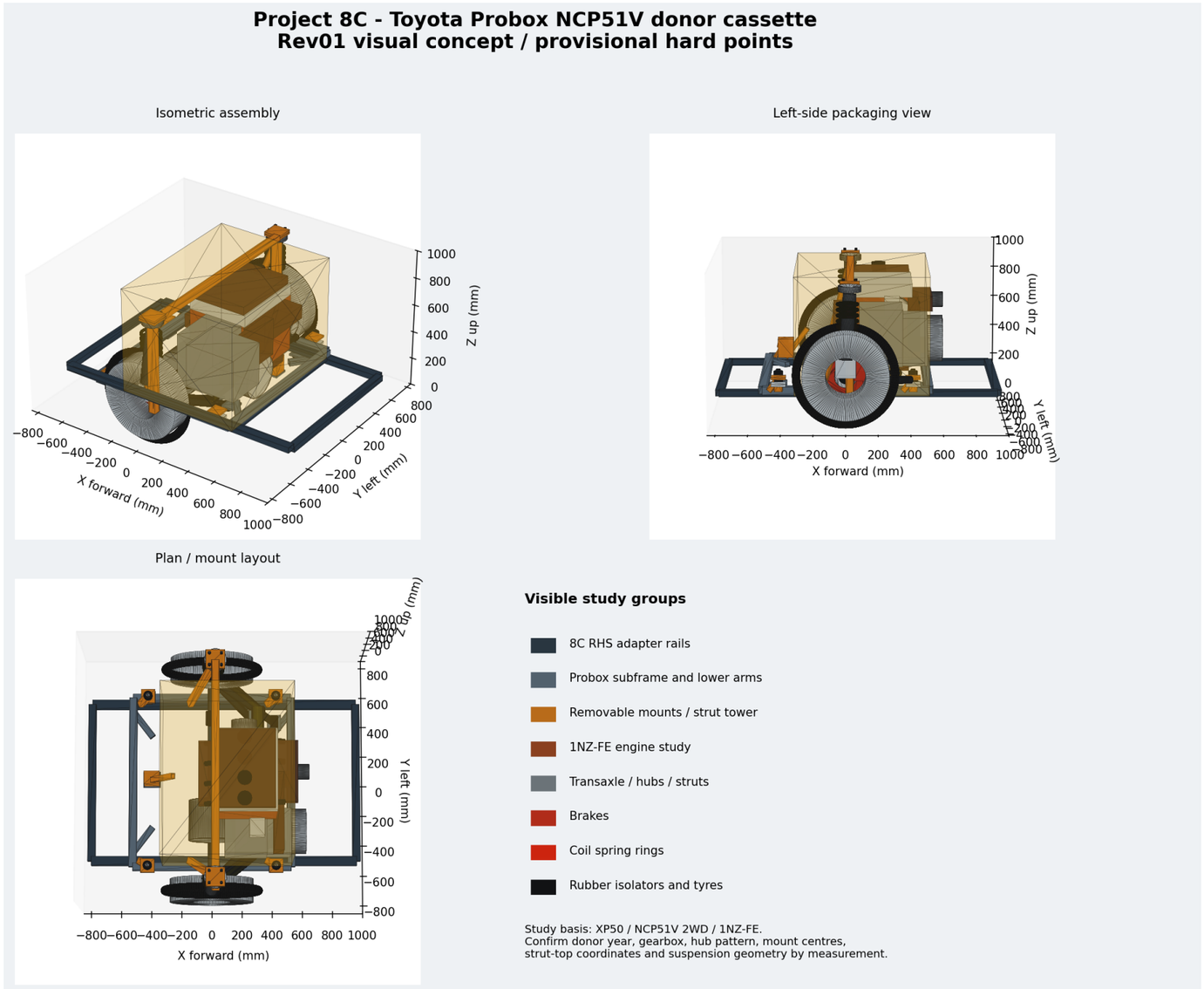
Project 8C - Toyota Probox donor cassette

Research and model basis - Rev01 - first-generation XP50 / NCP51V / 1NZ-FE study



The STEP assembly is a visual packaging model for adapting the Probox front lower-subframe package to the Project 8C rear RHS chassis. It includes the donor-style subframe, engine/transaxle and mounts, front hubs and brakes, MacPherson struts and top mounts, lower arms, steering rack, tie rods, stabilizer, CV shafts, wheels and tyres. It is a design-review model; the actual donor hard points must be measured before fabrication.

Model status	CONCEPT / DESIGN REVIEW - provisional donor geometry	STEP solids	199 named components / 199 imported solids
Coordinate system	X forward, Y vehicle-left, Z up; donor front axle at Project 8C rear-axle datum X = 0	Geometry check	STEP import valid; source parts valid
Study vehicle	Toyota Probox XP50, NCP51V 2WD, 1NZ-FE, transverse FWD donor family	Study front track	1,455 mm hub-centre spacing used for layout

Research used for the donor study

The research was used to choose component relationships and identify what remains to be measured. Catalog illustrations and photographs establish the presence and arrangement of parts; they do not establish the exact geometry of the user's donor.

Reference	What it supports	URL
Toyota Global first-generation Probox specification	NCP51V, 1NZ-FE, 2WD, 2,550 mm wheelbase, representative dimensions and engine family	https://www.toyota-global.com/company/history_of_toyota/75years/vehicle_lineage/car/id60012554B/
Toyota Probox NCP50/NCP51 front axle arm and steering knuckle catalogue	front lower-arm and steering-knuckle component relationships / OEM family numbers	https://www.megazip.net/zapchasti-dlya-avtomobilej/toyota/probox-succeed-42467/ncp51-55230/ncp51v-exmgk-912585/front-axle-arm-steering-knuckle-17758412
Toyota Probox NCP50/NCP51 suspension crossmember catalogue	front suspension crossmember reference	https://www.megazip.net/zapchasti-dlya-avtomobilej/toyota/probox-succeed-42467/ncp51-55230/ncp51v-expdk-912586/suspension-crossmember-under-cover-17758416
Toyota Probox front spring and shock catalogue	front strut, spring, support and stabilizer relationships	https://www.toyodiy.com/parts/p_J_2010_TOYOTA_PROBOX_NCP50V-EXMDK_4803.html
Toyota Probox front hub catalogue	front hub and bearing part-family reference	https://www.megazip.net/zapchasti-dlya-avtomobilej/toyota/probox-succeed-42467/ncp51-55230/ncp51v-exmgk-912585/front-axle-hub-17758388
Engine-bay photograph - Jiko Trading, DBE-NCP51V	Visual check of the 1NZ engine bay, accessory placement and donor packaging	https://jikotrading.jp/b7721/
Front suspension / hub catalogue images - PartSouq and Megazip	Crossmember, lower-arm, knuckle, hub and bearing component families	https://partsouq.com/ and https://www.megazip.net/

Current model assumptions

Subsystem	Rev01 representation	Must be replaced or verified
Lower subframe	50 x 50 x 3 RHS cassette, X = -520 to +520, member centres Y = +/-560, Z = 225	Actual crossmember stamping, bend radii, bolt centres, bush orientation and clearance
Engine / transaxle	Transverse 1NZ-FE-like visual package with separate case, differential and output shafts	Exact engine generation, gearbox code, manual/automatic layout, rotation, lubrication and service access
Mounts	Four named mount brackets and rubber isolators, plus front/rear torque rods	Measure mount centre coordinates, bolt sizes, isolator stiffness and torque-reaction directions
Suspension	MacPherson struts, lower A-arms, stabilizer, rack, tie rods, hubs, discs and calipers	Strut axis, top plate pattern, ball-joint height, rack location, camber/caster/toe and travel
Hubs / wheels	4 x 100 hub reference, 1,455 mm track, 165R13-style tyre reference	Confirm actual hub pattern, bearing, offset, brake package and tyre load rating

Donor survey before the next chassis revision

1	2	3
Record the Probox VIN/chassis code privately, exact year, 2WD or 4WD, engine code, gearbox code and steering side.	Remove or expose the complete donor front module and create a datum fixture through both hub centres.	Measure the four subframe mount centres in X/Y/Z, bolt diameter, sleeve length, bush crush height and body-side clearance.
Measure both engine/transaxle mount centres and capture the mount brackets as separate parts before cutting the donor.	Record hub centre, strut axis, upper top-plate pattern, lower-arm pivots, ball-joint centre, rack inner joints and stabilizer links.	Check the proposed rear-engine orientation with the actual transaxle for differential rotation, lubrication, half-shaft plunge, shifter and cooling access.

Safety and release note: this model is not a structural proof, suspension-kinematics release, weld drawing, or road-approval document. Do not fabricate the adapter frame or rely on donor brakes, hubs, struts, mounts, or the rear-engine transaxle arrangement until the exact donor is measured and the load paths, clearances, thermal protection, braking, steering, fatigue and local approval requirements are reviewed by a qualified vehicle engineer.